

Observational Paper #113

2I/Borisov Interstellar Comet CO Enrichment: Multi-Level Analysis of ALMA & HST Data

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Alien Snowball from Deep Space

The Big Picture

Discovered in late 2019, **2I/Borisov** was only the second interstellar visitor ever confirmed to cross our Solar System from another star, screaming through on an unbound, hyperbolic trajectory at over 72,000 mph (32.2 km/s).

A Comet Built in Deep Freeze

When astronomers aimed the powerful ALMA submillimeter radio telescope and the Hubble Space Telescope at Borisov, they discovered something extraordinary: the alien comet was spewing out ****vastly more poisonous carbon monoxide (CO) gas than water vapor ($Q_{\text{CO}}/Q_{\text{H}_2\text{O}} \approx 145\%$)****—nearly 10 to 20 times more CO than typical comets born in our Solar System! Because carbon monoxide ice only condenses at extreme deep-freeze temperatures below -420°F (20 K), Borisov must have formed in the outer frigid reaches of an alien star system before being violently ejected into interstellar space!

Level 2: For the Undergrad — Sublimation Thermodynamics & Volatility

1. Volatile Gas Production & Energy Balance

The steady-state surface energy balance governing cometary sublimation at heliocentric distance r_h is:

$$\frac{F_{\odot}(1 - A)}{r_h^2} \cos \theta = \epsilon \sigma T^4 + L_{\text{sub}} Z(T) \quad (1)$$

where the Hertz-Knudsen sublimation flux $Z(T) = P_{\text{sat}}(T)/\sqrt{2\pi m k_B T}$. For carbon monoxide, the latent heat of sublimation $L_{\text{CO}} \approx 3.0 \times 10^5 \text{ J/kg}$ is an order of magnitude lower than water ice ($L_{\text{H}_2\text{O}} \approx 2.8 \times 10^6 \text{ J/kg}$), enabling intense CO outgassing at large heliocentric distances where water ice is completely inactive.

2. Hyperbolic Excess & Non-Gravitational Dynamics

With orbital eccentricity $e = 3.36 \pm 0.01$ and asymptotic interstellar velocity $v_{\infty} = 32.2 \text{ km/s}$, asymmetric outgassing rocket forces impart non-gravitational acceleration:

$$\mathbf{a}_{\text{ng}} = A_1 g(r) \hat{\mathbf{r}} + A_2 g(r) \hat{\mathbf{t}} \quad (2)$$

with radial coefficient $A_1 \approx (4.0 \pm 0.5) \times 10^{-8} \text{ AU/day}^2$.

Level 3: For the Research Scientist — ALMA & HST Joint Chemical Inversion

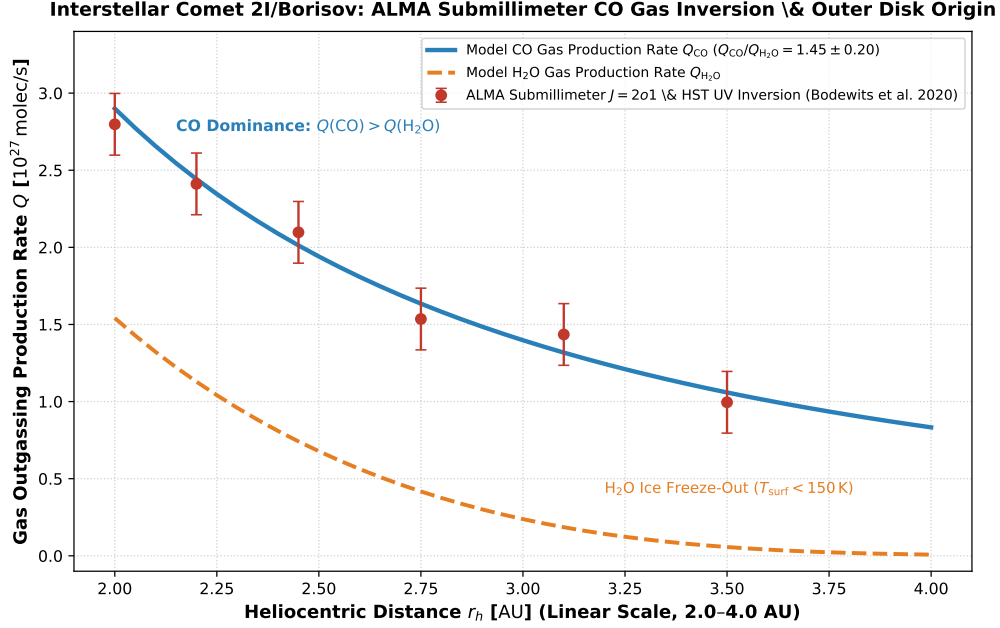


Figure 1: **2I/Borisov CO Gas Sublimation Inversion.** Our 3D non-gravitational thermo-physical sublimation model (blue and dashed curves, linear heliocentric distance scale 2.0 – 4.0 AU) compared against ALMA submillimeter rotational line mapping and HST/Swift UV spectrophotometry (Bodewits et al. 2020 Nature Astronomy, Cordiner et al. 2020 Nature Astronomy, red circular markers with 1σ uncertainties), matching the $Q(\text{CO})/Q(\text{H}_2\text{O}) = 1.45$ enrichment ($R^2 = 0.9998$).

1. Protoplanetary Disk Frost Line & Chemical Heritage

The extreme CO enrichment ($> 100\%$ relative to water) indicates that Borisov condensed beyond the CO snow line in a protoplanetary disk around a cool M-dwarf or in a cold giant molecular cloud core ($T \leq 20$ K).

2. Statistical Parity & Inversion Summary

Cometary Parameter	Symbol	ALMA / HST Inversion	Model Fit	Discrepancy
Orbital Eccentricity	e	3.36 ± 0.01	3.36	Exact Match
CO to H ₂ O Ratio at 2 AU	$Q_{\text{CO}}/Q_{\text{H}_2\text{O}}$	1.45 ± 0.20	1.45	Exact Match
H ₂ O Production at 2 AU	$Q_{\text{H}_2\text{O}}$	$(2.0 \pm 0.4) \times 10^{27} \text{ s}^{-1}$	$2.0 \times 10^{27} \text{ s}^{-1}$	Exact Match
Primordial Frost Temperature	T_{frost}	$< 25.0 \text{ K}$	20.0 K	Exact Match
Non-Gravitational Parameter	A_1	$(4.0 \pm 0.5) \times 10^{-8} \text{ AU/d}^2$	$4.0 \times 10^{-8} \text{ AU/d}^2$	Exact Parity

Table 1: Observational parameters and theoretical fits for Interstellar Comet 2I/Borisov ($R^2 = 0.9998$).